

- Hide menu
- Module 2 Introduction
- Lesson 1: Linear Regression and Least Squares
- Lesson 2: Modification of Linear Regression: Subset Selection
- Lesson 3: Coefficient Shrinkage for Linear Regression: Ridge Regression and LASSO
- Lesson 4: Data Transformations and Linear Regression
- Module 2 Summative Assessment
- Graded Assignment: Module 2 Summative Assessment
- Submitted
- Module 2 Summary

Your grade: 86.66%

Your score: 86.66% / Your highest score: 100.00%

Module 2 Summative Assessment

Next item →

Review Learning Objectives

Instructions

1. In which method is the risk of overfitting generally considered to be lower compared to best subset selection?

Least squares method

ⓧ

Ridge regression

Ⓢ

Forward stepwise selection

ⓧ

Best subset selection

ⓧ

Help me practice

Let's chat

0 / 1 point

Ready to review what you've learned before starting the assignment? I'm here to help.

Assignment details

Incorrect

Due

Attempts

Incorrect. Ridge regression shrinks the magnitude of the coefficients but does not push coefficients to zero to perform subset selection.

Submitted Nov 6, 12:04 AM CST

Try again

2. Why might a data scientist apply a square root transformation to a predictor in a linear regression model?

To ensure that all regression coefficients are positive.

ⓧ

To allow the regression to model non-linear relationships.

ⓧ

To increase the linearity of the relationship between predictor and response.

ⓧ

To reduce the effect of skewness and extreme values in the predictor.

Ⓢ

View submission

See feedback

3. What is a common technique to assess if linear regression adequately models the dependent variable?

Checking if the assumptions of linear regression are satisfied.

Ⓢ

Adding more predictors to the model until all p-values are significant.

ⓧ

Increasing the regularization parameter until the model's R-squared value peaks.

ⓧ

Using only categorical predictors to simplify the interpretation of coefficients.

ⓧ

Incorrect

Incorrect. Adding predictors indiscriminately can lead to overfitting and does not necessarily indicate model adequacy.

4. In the context of subset selection, what is a potential disadvantage of using forward stepwise selection over best subset selection?

It is more computationally intensive than best subset selection.

ⓧ

It might not include predictors that would be identified in a more exhaustive search.

Ⓢ

It requires that the number of predictors be less than the number of observations.

ⓧ

It evaluates too many models, leading to a high risk of overfitting.

ⓧ

Correct

5. In LASSO regression, what is the impact of the lambda penalty as it approaches zero?

The coefficients of the predictors are increasingly penalized and shrink to zero.

ⓧ

The model becomes equivalent to a Ridge Regression model.

ⓧ

The LASSO regression model becomes similar to an ordinary least squares regression model.

Ⓢ

The number of predictors selected by the model decreases.

ⓧ

Correct

6. Based on what you know regarding multicollinearity among predictors and its effect on ordinary least squares regression, how would the LASSO method handle multicollinearity by comparison?

It has no mechanism to handle multicollinearity, leading to unstable coefficients.

ⓧ

It amplifies the effect of multicollinearity when compared to linear regression on the coefficient estimates.

ⓧ

It can handle multicollinearity better because it shrinks coefficients of correlated predictors, sometimes to zero.

Ⓢ

It ignores multicollinearity entirely and is completely unaffected when compared to linear regression.

ⓧ

Correct

Correct: LASSO can deal with multicollinearity by shrinking the coefficients, potentially setting some to zero, thus reducing the variance of the estimates.

7. What does the slope in a simple linear regression model represent?

The maximum value of the dependent variable.

ⓧ

The rate at which the dependent variable changes for each unit change in the independent variable.

Ⓢ

The correlation coefficient between the dependent and independent variables.

ⓧ

The intercept of the regression line with the y-axis.

ⓧ

Correct

8. What is the effect of applying a regularization parameter that is too large in a Ridge Regression model?

It may lead to underfitting, where the model is too simplistic to capture the underlying pattern in the data.

Ⓢ

It will ensure that all predictor coefficients are set to zero.

ⓧ

It will cause the model to fit perfectly to the training data.

ⓧ

Correct

9. When is it appropriate to use a data transformation method over ordinary least squares regression?

When the data contain outliers or are heteroskedastic.

Ⓢ

When the data perfectly fits the assumptions of linear regression.

ⓧ

When the number of predictors is greater than the number of observations.

ⓧ

When the dependent variable is binary.

ⓧ

Correct

10. When dealing with a non-linear relationship between the independent and dependent variables, which transformation could be appropriate to attempt linearization?

Polynomial transformation.

Ⓢ

Absolute transformation

ⓧ

Binary transformation

ⓧ

Nominal transformation

ⓧ

Correct

11. In linear regression, what would adding interaction terms between predictors allow for?

It guarantees that the model will meet the assumption of multicollinearity.

ⓧ

It ensures that the residuals will be normally distributed.

ⓧ

It allows the model to represent the combined effect of predictors on the dependent variable.

Ⓢ

It simplifies the model by reducing the number of predictors.

ⓧ

Correct

12. What is the result of applying a differencing transformation to a time-series data in the context of linear regression?

It ensures that the future values of the series can be predicted without error.

ⓧ

It changes the time intervals of the data collection.

ⓧ

It converts the series into a stationary series.

ⓧ

It helps to remove trends and seasonality from the data.

Ⓢ

Correct

13. What is a common method used to select the regularization parameter in Ridge Regression and LASSO?

Always setting it to a default value of 1.

ⓧ

Cross-validation.

Ⓢ

Fixed rule based on the number of predictors.

ⓧ

Expert judgment only.

ⓧ

Correct

14. What effect does multicollinearity have on the estimates produced by a linear regression model?

It can make the coefficient estimates unstable.

Ⓢ

It leads to a decrease in the standard error of the regression coefficients.

ⓧ

It reduces the computational time required to fit the model.

ⓧ

It increases the overall fit of the model as measured by R-squared.

ⓧ

Correct

15. Why would a data scientist use a LASSO regression instead of traditional linear regression?

To guarantee that the model will have a higher R-squared value.

ⓧ

To allow the model to fit non-linear relationships without transformations.

ⓧ

To perform both variable selection and regularization to enhance prediction accuracy.

Ⓢ

To ensure that the residuals are normally distributed.

ⓧ

Correct

https://www.coursera.org/learn/linos-tech-statistical-learning/assignment-submission/lieh/module-2-summative-assessment/view-feedback

1/1